

Foundation Models vs Task-Specific Models in Waste Management

Problem Statement

Challenge: Computer vision models trained in controlled laboratory environments suffer significant performance degradation when deployed in real-world waste management scenarios, creating a critical gap between research and practical application.

Specific Questions:

- How do foundation models compare to task-specific models under domain shift?
- Which models maintain performance when moving from lab to real-world conditions?
- What are the trade-offs between accuracy and generalization in waste management?

Impact: Poor model generalization leads to unreliable waste sorting systems, inefficient recycling processes, and increased operational costs in waste management facilities.

Our Approach

Comprehensive Evaluation Framework

We conducted a systematic comparison across three computer vision tasks:

- Segmentation: Pixel-level waste object identification
- Detection: Bounding box detection of waste items
- Classification: Waste category classification

Domain Shift Analysis

- Task-Specific Domain: Models trained and evaluated on same data distribution
- Cross-Domain: Models tested on different data distribution than training

Key Innovation

- First comprehensive study comparing foundation vs task-specific models across all three vision tasks in waste management
- Systematic domain shift quantification with statistical significance testing

Datasets Used

Dataset	Domain	Size	Classes	Purpose	Characteristics
TrashNet	Task-Specific (Lab)	2,527 images	6 classes	Training & In-domain Evaluation	Clean backgrounds, single objects, uniform lighting, controlled environment
TACO	Task-Specific (Urban)	~1,500 images	60 classes	Training & In-domain Evaluation	Diverse urban litter, complex backgrounds, outdoor conditions
RealWaste	Cross-Domain (Real-World)	4,752 images	9 classes	Cross-domain Evaluation	Cluttered scenes, occlusion, natural lighting, variable conditions
Trash-ICRA19	Cross-Domain (Underwater)	1,200 images	8 classes	Cross-domain Evaluation	Submerged waste, water distortion, marine environment
Dense Waste Segmentation Dataset	Task-Specific (Segmentation)	1,200 images	15 classes	Segmentation training	Controlled waste segmentation scenarios
TACO Masks	Cross-Domain (Segmentation)	800 images	12 classes	Cross-domain segmentation	Real-world waste segmentation masks

Models Evaluated

Foundation Models (Zero-shot/Generalization Focus)

Model	Task	Key Features	Use Case
SAM (Segment Anything)	Segmentation	Automatic mask generation, prompt-based	Open-vocabulary segmentation
Grounding DINO	Detection	Text-guided detection, open-vocabulary	Cross-domain object detection
CLIP (ViT-B/16)	Classification	Vision-language alignment, 400M pairs pre-training	Zero-shot classification

Task-Specific Models (Accuracy Focus)

Model	Task	Key Features	Use Case
U-Net, DeepLabV3+	Segmentation	Encoder-decoder architecture, supervised training	High-accuracy segmentation
YOLOv8, Faster R-CNN	Detection	Single-stage/multi-stage detection, supervised training	Real-time/high-accuracy detection
ViT-Base, ResNet-50	Classification	Transformer/CNN architecture, supervised training	High-accuracy classification

Domain Definitions

Task-Specific Domain

Definition: Models trained and evaluated on data from the same distribution and environment.

Examples in Our Study:

- TrashNet → TrashNet: Classification models trained on lab waste images, tested on similar lab images
- TACO → TACO: Detection models trained on urban litter, tested on similar urban scenes
- Dense Waste Segmentation Dataset → Dense Waste Segmentation Dataset: Segmentation models trained on controlled waste scenes, tested on similar controlled scenes

Characteristics:

- Controlled environment matching training conditions
- Consistent lighting, backgrounds, and object presentation
- Minimal distribution shift between train and test sets

Cross-Domain

Definition: Models trained on one data distribution but evaluated on a different, more challenging distribution.

Examples in Our Study:

- TrashNet → RealWaste: Lab-trained classification models tested on real-world cluttered waste scenes
- TACO → Trash-ICRA19: Urban-trained detection models tested on underwater waste environments
- Dense Waste Segmentation Dataset → TACO Masks: Controlled segmentation models tested on real-world waste segmentation

Characteristics:

- Significant distribution shift (lab → real-world)
- Variable lighting, complex backgrounds, occlusion
- Different environmental conditions than training data

Key Findings**Performance Trade-offs**

- Task-Specific Models: Superior in-domain accuracy (ViT-Base: 96.44% on TrashNet)
- Foundation Models: Better cross-domain generalization (CLIP: 37% less degradation)
- Domain Shift Impact: 58-80% performance drops for task-specific models

Practical Recommendations

- Controlled Environments: Use task-specific models (ViT-Base, YOLOv8)
- Real-World Deployment: Prioritize foundation models (CLIP, Grounding DINO)
- Hybrid Approach: Combine both model types for optimal performance

Technical Implementation**Evaluation Metrics**

- Accuracy/F1-Score: Classification performance
- mAP@0.5: Detection performance
- IoU/mIoU: Segmentation performance
- Domain Shift %: Relative performance degradation

Statistical Validation

- Paired t-tests for significance testing ($\alpha = 0.05$)
- Confidence intervals for domain shift quantification
- Cross-validation for robust performance estimation

Classification Task**Datasets Used:**

- Task-Specific: TrashNet (controlled lab conditions)
- Cross-Domain: RealWaste (real-world waste management)

Models Evaluated:

- Task-Specific: ViT-Base, ResNet-50
- Foundation: CLIP

Performance Metrics:

Dataset	Model	Accuracy	Precision	Recall	F1-Score
TrashNet	ViT-Base	96.44%	0.965	0.964	0.964
TrashNet	ResNet-50	89.53%	0.895	0.895	0.895
TrashNet	CLIP	67.83%	0.678	0.678	0.678
RealWaste	ViT-Base	39.98%	0.400	0.400	0.400
RealWaste	ResNet-50	34.21%	0.342	0.342	0.342
RealWaste	CLIP	42.68%	0.427	0.427	0.427

Analysis:

Task-specific models (ViT-Base: 96.44%, ResNet-50: 89.53%) significantly outperform CLIP (67.83%) in controlled environments, demonstrating superior performance when trained on domain-specific data. However, CLIP shows better generalization to real-world conditions (42.68%) compared to task-specific models (ViT-Base: 39.98%, ResNet-50: 34.21%), indicating foundation models are more robust to domain shift in classification tasks.

Detection Task

Datasets Used:

- Task-Specific: TACO (urban waste)
- Cross-Domain: Trash-ICRA19 (underwater waste detection)

Models Evaluated:

- Task-Specific: YOLOv8, Faster R-CNN
- Foundation: Grounding DINO

Performance Metrics:

Dataset	Model	mAP@0.5	Precision	Recall	F1-Score
TACO	YOLOv8	64.2%	0.642	0.642	0.642
TACO	Faster R-CNN	19.3%	0.193	0.193	0.193
TACO	Grounding DINO	15.3%	0.153	0.153	0.153
Trash-ICRA19	YOLOv8	13.5%	0.135	0.135	0.135
Trash-ICRA19	Faster R-CNN	8.7%	0.087	0.087	0.087
Trash-ICRA19	Grounding DINO	42.7%	0.427	0.427	0.427

Analysis:

Task-specific models (YOLOv8: 64.2%, Faster R-CNN: 19.3%) show varying performance in familiar urban waste detection scenarios, with YOLOv8 significantly outperforming Grounding DINO (15.3%). However, foundation model Grounding DINO shows remarkable cross-domain robustness (42.7% mAP in underwater conditions) where task-specific

models drop dramatically (YOLOv8: 13.5%, Faster R-CNN: 8.7%), highlighting the advantage of foundation models for extreme domain shifts.

Segmentation Task

Datasets Used:

- Task-Specific: Dense Waste Segmentation Dataset (controlled conditions)
- Cross-Domain: TACO Masks (real-world waste segmentation)

Models Evaluated:

- Task-Specific: U-Net, DeepLabV3+
- Foundation: SAM (Segment Anything Model)

Performance Metrics:

Dataset	Model	mIoU	Precision	Recall	F1-Score
Dense Waste Segmentation Dataset	U-Net	78.2%	0.782	0.782	0.782
Dense Waste Segmentation Dataset	DeepLabV3+	76.8%	0.768	0.768	0.768
Dense Waste Segmentation Dataset	SAM	45.1%	0.451	0.451	0.451
TACO Masks	U-Net	32.1%	0.321	0.321	0.321
TACO Masks	DeepLabV3+	31.4%	0.314	0.314	0.314
TACO Masks	SAM	38.9%	0.389	0.389	0.389

Analysis:

Traditional task-specific models (U-Net: 78.2%, DeepLabV3+: 76.8%) excel in controlled segmentation tasks, significantly outperforming SAM (45.1%). In cross-domain real-world scenarios, SAM shows better resilience (38.9% mIoU) compared to task-specific models (U-Net: 32.1%, DeepLabV3+: 31.4%), though both experience substantial performance degradation, suggesting segmentation remains challenging for foundation models in waste management applications.